R22

Code No.: 22CSC15N

CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.E. / B.Tech (CSE/CET) V Sem (Main) Examination December 2024

Operating Systems

Time: 3 Hours **Max Marks: 60**

Note: Answer **ALL** questions from **Part-A** at one place in the same order and **Part–B (Internal Choice)**

Part - A

(5Q X 2M = 10 Marks)

			M	CO	BT
1		What is the main difficulty that a programmer must overcome in writing an operating system for a real-time environment?	(2)	1	1
2		Describe the Micro-kernel and Macro-Kernel Approaches of Operating Systems with example. Write their causes.	(2)	2	2
3		How does paging eliminate external fragmentation?	(2)	3	1
4		What is the difference between Dekker's solution and Peterson's solution for two-process synchronization solution?	(2)	4	1
5		List the security attacks that cannot be prevented by encryption.	(2)	5	1

Part - B

(5Q X 10M = 50 Marks)

			M	CO	BT
6	(a)	The kernel of an OS masks off all interrupts during interrupt servicing. Discuss the advantages and disadvantages of such masking.	(5)	1	2
	(b)	Multi-programming is inherent in multiuser and multitasking systems. Explain how.	(5)	1	2
		(OR)			
7	(a)	What are the functions of an OS from user's viewpoint and system's viewpoint? Explain.	(5)	1	2
	(b)	Describe the Micro-kernel and Macro-Kernel Approaches of Operating Systems with example.	(5)	1	2
8	(a)	What is the need of multi-level queue scheduling? Explain the drawback in implementing multi-level queue scheduling?	(5)	2	2
	(b)	Consider the following scenario of processes with quantum time = 2. Draw the Gantt chart for the execution of the processes, showing their start time and end time using FCFS, and Round Robin. Calculate average turnaround time and average waiting time for the system.	(5)	2	3

Process	Arrival time	Execution time
P1	0	5
P2	1	9
P3	2	7
P4	3	2
P5	4	4

(OR)

9	(a)	A parent process needs to be terminated but has two child processes: One child process is waiting for an I/O, and the other is executing. How will you terminate the parent process? Explain.	(5)	2	3
	(b)	In a system, Process A is running, and Processes B and C are in a blocked state. An I/O-completion signal arrives for Process B. The priority of Process B is the highest. Describe the sequence of actions in the operating system.	(5)	2	3
10	(a)	Discuss the address translation in the two-level page table structure.	(5)	3	2
	(b)	A process has relocatable code of size 700 K. The relocation register is loaded with 30010 K and the limit register contains the address 31000 K. If the processor generates logical addresses, 990 and 1020, where will they be located in the physical memory?	(5)	3	3
		(OR)			
11	(a)	Show the stack implementation of LRU on the following string: 1 0 5 1 1 3 5 1 5 3 4 5 2 1 3 0 1 4 0 5	(5)	3	3
	(b)	C-LOOK is an improvement over LOOK scheduling algorithm. However, it may lead to increased average response time and lower throughput. Explain how.	(5)	3	3
12	(a)	What is a semaphore? Explain its initial implementation.	(5)	4	2
	(b)	Explain the implementation of kernel threads.	(5)	4	2
		(OR)			
13	(a)	Prove that circular wait condition may be produced, if all the other three necessary conditions of a deadlock are present in a system.	(5)	4	2
	(b)	Explain the deadlock and starvation problems in dining-philosophers' problem.	(5)	4	2
14	(a)	A file named data is frequently accessed by users in a system. The following alternatives are proposed to simplify access to data. **a.** Set up links from every user's home directory to data. **b.** Copy data into every user's home directory. Compare the advantages and drawbacks of these approaches.	(5)	4	3
	(b)	Describe the File System Actions at *close* with an illustrative example.	(5)	4	2
		(OR)			
15	(a)	Explain the malware taxonomy with a detailed description.	(5)	4	2
	(b)	Provide a detailed description on OS-level memory protection.	(5)	4	2

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